

MANUAL FOR USING THE WALLTOPIA PRELIMINARY LOADS CALCULATOR

PRELIMINARY LOADS - NOT FOR CONSTRUCTION

Web and mobile workflow | Metric and imperial units

I. Purpose and scope

The Walltopia Preliminary Loads Calculator is a multiplatform tool for selecting and presenting characteristic preliminary loads imposed by artificial climbing structures (ACS) on an existing building structure.

It reproduces the approved Walltopia preliminary-load datasets through a guided interface and presents the selected values in tables and technical diagrams.

IMPORTANT

The calculator is intended for **preliminary assessment only**. It **does not** perform final structural design, anchor design, connection verification or code compliance verification.

The calculator supports:

- Climbing walls with protection points and bouldering walls.
- Metric (kN, m) and imperial (lb, ft) unit systems.
- Selection of wall height, attachment scheme, column span, overhang and live-load force level.
- Characteristic and optional factored values, capacity orientation, 2D diagrams, PDF reports, saved projects and Engineering Support inquiries.

Load values shall be reviewed and factored by a qualified structural engineer according to the applicable national standard.

III. Worked example

III.1. Example 1 - climbing wall, metric units

1. Choose the preferred units

Select **Metric (kN, m)** for characteristic forces in kilonewtons and dimensions in metres. Select **Imperial (lb, ft)** when the project requires US customary units.

2. Choose the structure type

Select **Climbing wall** for an ACS with protection points or **Boulder wall** for a structure without protection points. The applicable heights, attachment schemes and load datasets are different.

INPUTS

Units

Metric (kN, m)
Imperial (lbs, ft)

Structure type

Climbing wall
Bouldering wall

Wall provision point: EN 12572 / ACS with partial safety factor ⓘ

Climbing wall height (m)

6
9
10
12
15
18
21
24

Attachment scheme
Center of attachment by height (m)

2
4
6
8
10

Overhang angle (°)

-15
-10
-5
0
5
10
15

Check against your structure (optional) ⓘ

Allowable column axial

150.00
kN

Horizontal load (wind acting column / frame)

15.50
kN

☐ Show footnote (if any)

CALCULATE →

3. Select the climbing-surface height

Choose the nominal height of the climbing surface. Only heights covered by the approved preliminary-load dataset are available.

For this worked example select 12 m.

4. Select the attachment scheme

The attachment scheme defines the number and elevation of attachment levels along the height of the ACS. Available schemes are filtered automatically after the wall height is selected.

For this example select 3 attachment levels: Z1 = 4.0 m, Z2 = 8.0 m and Z3 = 11.5 m.

INPUTS

Units

Metric (kN, m)Imperial (lbs, ft)

Structure type

Climbing wallBouldering wall

Wall provision point: EN 12572 / ACS with partial safety factor ☐

Climbing wall height (m)

6910**12**15182124

Attachment scheme

Center of attachment by height (m)

24**6**810

Overhang angle (°)

-15-10-5**0**51015

Check against your structure (optional) ⓘ

Allowable column axial150.00kN

Horizontal load (wind acting column / frame)15.50kN

☐ Show footnote (if any)

CALCULATE→

5. Select column span A

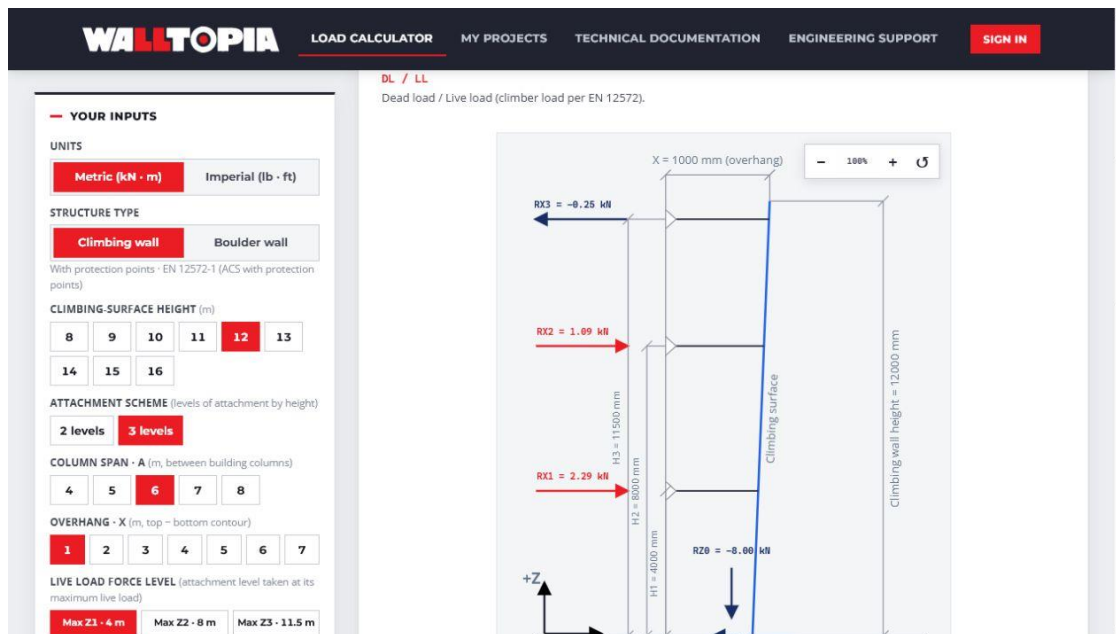
A is the span between the existing building columns or frames. The selected span affects the loads reported for the building columns, especially the dead-load components.

For this example select **A = 6 m**.

6. Select overhang X

X is the horizontal difference between the bottom and top contours of the climbing surface. Use the representative or most unfavourable overhang in the vicinity of the investigated column or frame.

For this example select **X = 1 m**.



7. Select the live-load force level

The force-level control selects the attachment level at which the live load is taken at its local maximum. The concurrent live-load values shown at the other levels belong to the same load situation.

Review each available force level separately when identifying the governing action for an existing column or frame.

Climbing wall 12 m

2 attachment walls, span 4 m, 6 m overhang, 2.4 m characteristic values in kN

Governing column load

23.17 kN (factored)

FORCE LEVEL – ATTACHMENT LEVEL TAKEN AT ITS MAXIMUM LIVE LOAD

Level 1 (4 m)

Level 2 (8 m)

Level 3 (11.5 m)

ATTACHMENT AT NET	R - 2.4 (min. characteristic)	R - 1.0 (mean characteristic)	R - 0.5 (max. characteristic)	R - 1.6L (max. characteristic)
Base — vertical (Z)	-5.02	-3.00	—	—
Base — horizontal (X)	+1.00	+2.00	—	—
Level 1 (4 m)	+0.56	2.00	4.10	7.15
Level 2 (8 m)	+0.46	1.67	6.64	2.96
Level 3 (11.5 m)	+0.29	+1.29	5.28	+0.69

COORDINATE SYSTEM & SYMBOLS

Z

NOTES

IMPORTANT

Do not combine the maximum live-load value from every force-level selection into one simultaneous load case. Use **one selected force-level situation at a time** with its displayed concurrent values.

IV. Reading the results

The results table separates loads on one ACS axis from the loads applied to the existing building column or frame.

Symbol	Meaning	Use
R	Load on one axis of the ACS	Base points and single-axis attachment points
L	Load on a building column / frame	Primary values for assessment of the existing structure
DL	Dead load	Acts on the supporting structure and must be factored
LL	Live load	Climber load case with the selected maximum force level
Z0	Vertical component at the base	Vertical load at the ACS base
X0-X3	Horizontal components	Base and attachment-level actions

Climbing wall 12 m

2 attachment walls, span 4 m, 6 m overhang, 2.4 m characteristic values in kN

Governing column load

23.17 kN (factored)

FORCE LEVEL – ATTACHMENT LEVEL TAKEN AT ITS MAXIMUM LIVE LOAD

Level 1 (4 m)

Level 2 (8 m)

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ATTACHMENT AT NET	R - 2.4 (min. characteristic)	R - 1.0 (mean characteristic)	R - 0.5 (max. characteristic)	R - 1.6L (max. characteristic)
Base — vertical (Z)	-5.02	-3.00	—	—
Base — horizontal (X)	+1.00	+2.00	—	—
Level 1 (4 m)	+0.56	2.00	4.10	7.15
Level 2 (8 m)	+0.46	1.67	6.64	2.96
Level 3 (11.5 m)	+0.29	+1.29	5.28	+0.69

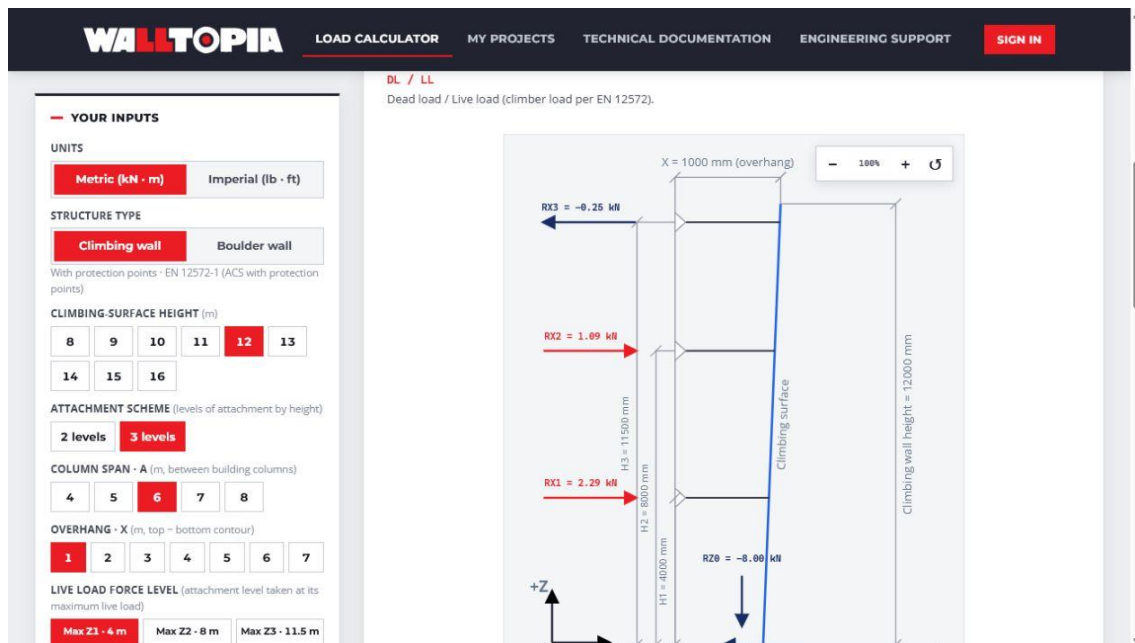
COORDINATE SYSTEM & SYMBOLS

Z

NOTES

V. Technical 2D diagram

The diagram illustrates the selected ACS geometry, the existing support line, attachment elevations and the direction and magnitude of the selected load components.



Red arrows identify the live-load components for the selected force-level situation. Blue or dark arrows identify other load components. Positive and negative signs follow the coordinate system shown in the diagram.

VI. Check against the existing structure

The optional capacity field provides a quick orientation by comparing the governing calculated column load against a user-entered allowable value.

Enter the allowable column load supplied by the responsible structural engineer, then run the capacity check. The calculator reports whether the entered capacity is exceeded by the governing value.

INPUTS

Units

Metric (kN, m)
Imperial (lbs, ft)

Structure type

Climbing wall
Bouldering wall

Wall provision point: EN 12572 / ACS with partial safety factor ⓘ

Climbing wall height (m)

6
9
10
12
15
18
21
24

Attachment scheme
Center of attachment by height (m)

2
4
6
8
10

Overhang angle (°)

-15
-10
-5
0
5
10
15

Check against your structure (optional) ⓘ

Allowable column axial

150.00
kN

Horizontal load (wind acting column / frame)

15.50
kN

☐ Show footnote (if any)

CALCULATE →

IMPORTANT

A passing indication is not a structural verification. The comparison does not assess stability, combined actions, connections, anchors, local effects, fatigue, seismic effects or the complete set of applicable load combinations.

VII. Characteristic and factored values

The approved load tables contain characteristic preliminary values. Where the interface offers factored values, the displayed factors are an orientation based on the selected regional dataset.

For the metric EU workflow, the interface may present the commonly used factors 1.35 for dead load and 1.50 for live load. For the USA workflow, the paired factors are 1.20 and 1.60. The responsible engineer must confirm the governing standard, load combinations and project-specific factors.

IMPORTANT

Never treat the optional factored display as a substitute for the structural design load combinations required for the project.

Ambiguous signs

Some load entries may contain positive and negative alternatives because different climbing-surface geometries with the same nominal input can produce different governing directions. The designer shall evaluate the unfavourable sign for the investigated structural system.

VIII. Standard attachment details

The attachment configurator helps the user browse standard details after choosing the supporting slab and the attachment method. The available choices are filtered by the selected ACS type and support category.

The user may select details for a solid concrete slab or hollow-panel slab and review applicable support categories such as a concrete wall or column, steel column, and - where available - masonry wall.

Concrete floor

Concrete wall

Steel column

Masonry wall

REVISIONS	
No.	Date

CLIENT: _____

Country - City - Name _____

DRAWING INCLUDES
Details

DRAWING ENGINE: [blank]
CHECKED ENGINE: [blank]
SCALE: 1:10 (mm) A1
PROJECT NO.: [blank]
SHEET: 1/10
DRAWING: [blank]

IMPORTANT

The configurator does not size anchors or verify the resistance of a connection. Every selected detail must be checked for the actual substrate, geometry, loads and applicable standard.

IX. Saving projects and exporting reports

Signed-in users can save a calculation as a project, add tags and custom properties, search the project dashboard, reopen a previous calculation and save an updated copy.

The PDF export records the selected inputs, calculated preliminary loads, key notes and technical diagrams. Store the report together with the project revision used for the structural review.

IMPORTANT

Before issuing a report, confirm the units, ACS type, height, attachment scheme, span A, overhang X and force level. A PDF only records the selected calculation state; it does not validate the input data.

Recommended project record

- Project name and revision.
- Building and location.
- Selected ACS geometry and supporting frame.
- Calculation date and responsible user.
- All force-level situations reviewed.
- Exported PDF and any Engineering Support correspondence.

X. Engineering Support and mobile use

A saved project can be linked to a technical inquiry. The Engineering Support form includes the project calculation summary so the reviewing engineer can identify the selected geometry and load case.

Describe the existing structure, the issue requiring review and the exact decision or clarification needed. Responses are sent by email.

PROJECT

Climbing wall 12 m

3 attachment walls • span 8 m • overhang 3 x 2 m
Characteristic values in kN

Governing column load
23.17 kN (factored)

UNITS
Metric (kN, m) Imperial (lb, ft)

STRUCTURE TYPE
Climbing wall Boulder wall

With protection points: EN 12572-1 ACS with protection points

CLIMBING WALL HEIGHT (m)

6	9	10	11	12
13	14	15	16	

ATTACHMENT SCHEME (levels)

2 levels 3 levels

COLUMN SPAN - A (m)

4	5	6	7	8
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OVERHANG - X (m)

-1	0	1	2	3	4
5					

CHECK AGAINST YOUR STRUCTURE (optional)

Allowable column load 150.00 kN

Horizontal load (wind acting column / frame) 15.50 kN

☐ Show footnotes (if any)

CALCULATE

RESULTS DIAGRAM NOTES

Governing column load
23.17 kN (factored)

FORCE LEVEL - ATTACHMENT LEVEL TAKEN AT ITS MAXIMUM LIVE LOAD

	Level 1 (4 m)	Level 2 (8 m)	Level 3 (12.5 m)
Base - vertical (Z)	-5.02	-3.00	-
Base - horizontal (X)	+1.00	+2.00	-
Level 1 (4 m)	+0.56	2.00	4.10
Level 2 (8 m)	+0.48	1.67	6.64
Level 3 (12.5 m)	+0.29	1.29	5.28

COORDINATE SYSTEM & SYMBOLS

Load on one axis of the ACS - base points and single-axis attachment points.

Load on a building column / frame, with the column span A taken into account. Focus on these for the existing structure.

Vertical component at the base
X0 / X1 / X2 / X3

Horizontal component at the base (X0) and at attachment level 1, 2, 3

DL / LL
Dead load / Live load (climber load per EN 12572)

VIEW NOTES

On mobile devices the same calculation inputs and results are presented in a vertical layout. Review the Results, Diagram and Notes sections before saving or exporting the project.

XI. Final calculation checklist

Before using the preliminary loads in an engineering assessment, confirm all items below:

- Correct metric or imperial unit system selected.
- Correct structure type selected: climbing wall or bouldering wall.
- Wall height and attachment scheme match the project geometry.
- Column span A represents the investigated existing frame.
- Overhang X represents the local or most unfavourable geometry.
- Every applicable live-load force level has been reviewed separately.
- Signs and coordinate directions have been interpreted correctly.
- Characteristic values have been factored according to the governing standard.
- Special conditions at end and corner columns have been considered.
- Attachment details and anchors will be verified by a qualified engineer.
- The exported report is marked PRELIMINARY LOADS - NOT FOR CONSTRUCTION.

IMPORTANT

For anything beyond preliminary sizing or where the input geometry is outside the calculator dataset, contact Walltopia Engineering Support.